

Finite-depth Lyapunov-growth estimation in GNNs

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Status: reproducible pilot study, 2026-09-24.

Question: how should we estimate the top depth-wise Jacobian growth rate of a trained GNN, and how dependable are random tangent probes across GNNs and graph/data conditions?

This report concerns the finite network quantity

$$\lambda_K^{\text{top}}(G, x; \Phi) = \frac{1}{K} \log \sigma_1(B_K), \quad B_K = D_{h_0}(F_{K-1,G} \circ \dots \circ F_{0,G})(h_0),$$

where h_0 is the output of the encoder and h_K the output of the last hidden propagation layer. It does **not** include encoder or decoder derivatives and is not automatically an infinite-depth Lyapunov exponent. All results use the complete layer derivative at the forward state, including activation and state-conditioned sheaf restriction-map derivatives.

Primary-estimand implementation detail: no tangent vector is mean-centered or projected between layers when estimating B_K . Pierre’s P -projected procedure measures a different product, so it is evaluated and labeled separately below. Scalar norm renormalization is used only to prevent numerical overflow/underflow.

Executive conclusions

1. **A maximum over independent JVP probes is correct but was not useful as a point estimate here.** It is a rigorous lower bound for every finite M and converges almost surely as $M \rightarrow \infty$. Yet even **1,024** probes did not meet the predeclared 0.01-per-layer error target in any of the 432 trained calibration streams: pooled median gaps remained 0.395 for GCN and 0.109 for the deep sheaf baseline.
2. **Matrix-free power iteration on $B_K^* B_K$ is the right default for the top finite-depth rate when VJPs are available.** With three starts and 160 iterations, all 36 tiny trained calibration cells met residual $\leq 10^{-7}$ and agreed with reverse-mode dense-SVD references to mean absolute rate error 3.9×10^{-17} (maximum 2.2×10^{-16}).
3. **Probe count is architecture/spectrum dependent, not a universal GNN hyperparameter.** In this study, random probes were looser for the depth-3 GCN and depth-3 ChebNet than for the depth-10 sheaf baseline despite comparable tangent dimensions; graph family had a smaller effect than architecture. Training did not make random probes reliably tighter in the initialization ablation.
4. **Do not infer oversquashing or task utility from this top rate alone.** The GCN and sheaf baseline have positive top hidden-state rates here while their test accuracy differs sharply under a heterophilic distribution shift. Top global sensitivity does not identify source-to-target communication or spatial delocalization.
5. **Pierre’s mean-zero projection gives a smaller, deliberately different observable.** Matched projected-cocycle exact rates averaged 0.344 rather than 0.674 for GCN, 0.427 rather than 0.523 for sheaf, and 0.844 rather than 1.112 for ChebNet. This supports the interpretation that a full-space top rate can be elevated by consensus-like modes, while also showing that non-constant modes still grow on average in these trained examples.

The findings are strong evidence for estimator choice on these controlled models, not a claim that all GNNs or real-world datasets behave identically. Broader datasets, architectures and hyperparameter budgets are the next research phase.

Estimators: what each algorithm estimates

Algorithm	Quantity	What is rigorous	Practical verdict
Dense Jacobian + SVD	Exact λ_K^{top}	Exact finite- K reference	Use only at tiny tangent dimension
M full-space JVP probes	$\hat{\lambda}_{K,M}$	Finite- M lower bound; a.s. convergence as $M \rightarrow \infty$	Use as a lower-bound/spectral-shape diagnostic, not the default point estimate
JVP/VJP power on $B_K^* B_K$	Approximation to λ_K^{top}	Standard power convergence conditional on start overlap and spectral separation; final residual is directly checkable	Default top-rate estimator
Lanczos/Krylov on $B_K^* B_K$ (matrix-free)	Approximation to the same top singular value	Matrix-free extremal-eigenvalue method; largest Ritz value + Ritz residual	Default scalable route; **5-10 $\times$ faster than 3 $\times$ 160powerinbench* * QR/Benettinframe Long-depthLyapunovspectrumunde spectrummetho Notintercha singular values
Layerwise singular norms/product bounds Finite nonlinear perturbations	An upper bound or a different diagnostic Secant growth	Not $\sigma_1(B_K)$ in general Depends on perturbation scale and activation region	Never report as this exponent Optional autodiff audit only

Here $\hat{\lambda}_{K,M} = \max_{m \leq M} K^{-1} \log \|B_K q_m\|$. The key random-probe obstruction is visible in the rank-one case. If $B = \sigma uv^\top$ and q is uniform on S^{D-1} , then $\|Bq\|/\sigma = |v^\top q|$ and $(v^\top q)^2 \sim \text{Beta}(1/2, (D-1)/2)$. Typical overlap is $O(D^{-1/2})$. Thus a narrow leading direction can require very many parallel probes even though a power step aligns it immediately. A broad near-top singular subspace produces the opposite behavior. This is why a probe budget cannot be fixed from depth alone.

Controlled GNN architectures

Exact equations and scope are in notes/experiment-baselines.md. We used:

- **GCN**: a depth-3 Kipf-Welling stack, $h_{\ell+1} = \text{ReLU}(\tilde{A}h_\ell W_\ell)$, with self-looped symmetric normalized adjacency.
- **ChebNet**: depth 3 and Chebyshev order 3, with the self-loop-free symmetric normalized Laplacian and $\lambda_{\max} = 2$ scaling; it supplies a polynomial-filter contrast to one-hop GCN propagation.
- **Deep feature-conditioned diagonal-sheaf baseline**: depth 10, endpoint-dependent diagonal restriction maps, an assembled sheaf-Laplacian action and a residual ReLU update. It is inspired by Bodnar et al.'s neural sheaf diffusion, but is **not** claimed to reproduce every detail of that paper. Its implemented normalization is one-sided row normalization, not symmetric $D^{-1/2} L_{\mathcal{F}} D^{-1/2}$.

Datasets, splits, and the input-dependent rate

Graphs, features, and labels

This is a supervised node-classification pilot using deterministic synthetic graph generators. Every feature matrix has four columns. For node label $y_i \in \{0, 1\}$, the input signal is

$$x_i = (2y_i - 1)e_1 + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I_4).$$

Thus the first coordinate is a noisy class signal and the other three coordinates are noise; the configured noise standard deviation is 1.0. All node labels are used in the graph-level cross-entropy training loss.

- **Homophilic contextual SBM:** 12 nodes, with six nodes in each class. Undirected edges are sampled independently in the upper triangle with within-class probability $p_{\text{same}} = 0.75$ and between-class probability $p_{\text{other}} = 0.10$.
- **Heterophilic contextual SBM:** the same 12-node balanced classes and feature model, but the probabilities are reversed: $p_{\text{same}} = 0.10$ and $p_{\text{other}} = 0.75$. This is a structural distribution shift; it is never used for training or hyperparameter selection.
- **Barbell:** two six-node cliques joined by three intermediate path nodes, hence 15 nodes total. Labels divide the graph into left and right sides, with the class boundary inside the bridge. Features use the same noisy class signal. This is a bottleneck/topology shift and is also test-only.

The stored adjacencies contain no self-loops. Each architecture then applies its documented normalization: the GCN adds self-loops before symmetric normalization, whereas ChebNet uses the self-loop-free normalized Laplacian. The sheaf baseline constructs its own learned sheaf operator from the stored graph.

Training, validation, testing, and calibration

The split consists of **16 homophilic training graphs, 6 independently generated homophilic validation graphs, and 12 independently generated test graphs for each of the three conditions** (36 test graphs total). The base data seed is 101: training uses seeds 101–116, validation 10101–10106, and the homophilic, heterophilic, and barbell test generators start at 20101, 20102, and 20103 respectively. Exact per-example seeds and generator conditions are recoverable from the committed config and manifest. The generated examples are fixed across architectures and weight-initialization seeds.

Each hyperparameter candidate is trained for 120 full-graph Adam epochs with weight decay 10^{-4} ; the loss is the mean of the 16 per-graph node-classification losses. For each of two model seeds (11 and 29), validation accuracy alone selects hidden width from $\{8, 12\}$ and learning rate from $\{0.003, 0.01\}$. Test data do not enter model or hyperparameter selection. Reported test accuracy uses all 12 graphs in each condition. Jacobian calibration is more expensive and therefore uses the first **three** test examples per condition, giving 2 model seeds $\times$ 3 conditions $\times$ 3 examples = **18 pointwise calibration cells per architecture**. Each unprojected cell has an independent dense reference, a residual-checked power estimate, and 12 nested random-probe streams.

What varies when $\lambda_K^{\text{top}}(G, x; \Phi)$ is reported

The rate is evaluated **pointwise**. Within one calibration cell, the graph G , input feature signal x , trained parameters Φ , stochastic state, and resulting forward trajectory $h_0, \dots, h_K$ are fixed. The random vectors q_m sample tangent directions at that fixed h_0 ; they are not new data inputs. Every J_ℓ is evaluated along that input-specific trajectory. Consequently, the almost-sure $M \rightarrow \infty$ statement for random probes and the dense/power comparison both hold separately for each fixed (G, x, Φ) .

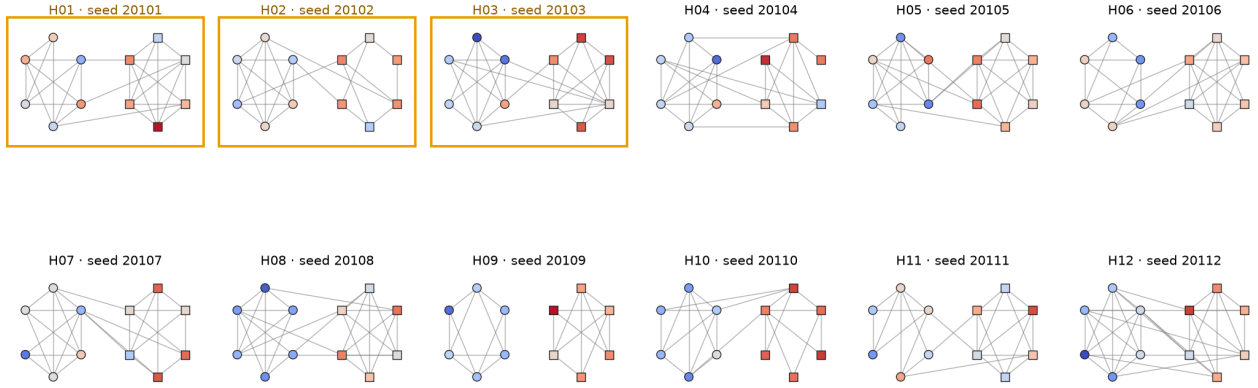
Across cells, the report treats $\lambda_K^{\text{top}}(G, x; \Phi)$ as an empirical random variable: boxplots and aggregate gaps pool the pointwise values from held-out examples, graph conditions, and the two trained model seeds. This pilot does **not** isolate dependence on x at fixed G : each synthetic draw generates the graph and its feature noise jointly, so within-condition spread conflates graph-to-graph variation, input-signal variation, and model-seed variation. With only three calibration examples per condition and seed, the displayed distributions are descriptive rather than population-level estimates. Also, h_0 is the encoded input and the reported derivative excludes encoder/readout derivatives; it is not the end-to-end $D_x \Phi_G(x)$.

The 36 held-out test graph/signal examples

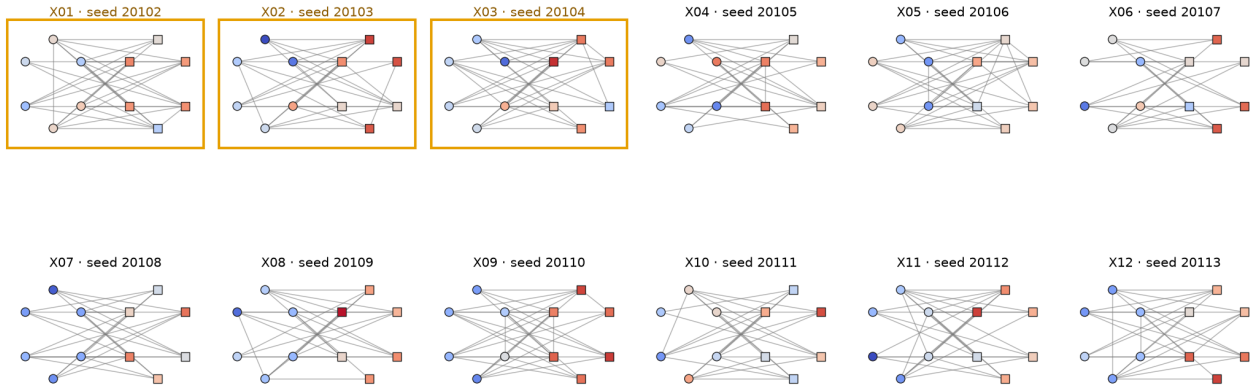
The complete split has 58 unique graphs: 16 training, 6 validation, and 36 testing. The next page shows all **36 test graphs**. Colour gives x_{i1} ; circles/squares give labels 0/1; the other three Gaussian-noise coordinates are omitted. Gold frames mark the **nine unique (G, x) calibration examples** (first three per family), reused across model seeds and architectures.

All 36 held-out test graph/signal examples

Homophilic SBM



Heterophilic SBM



Barbell

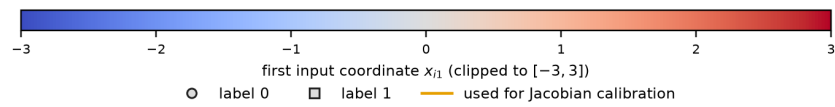
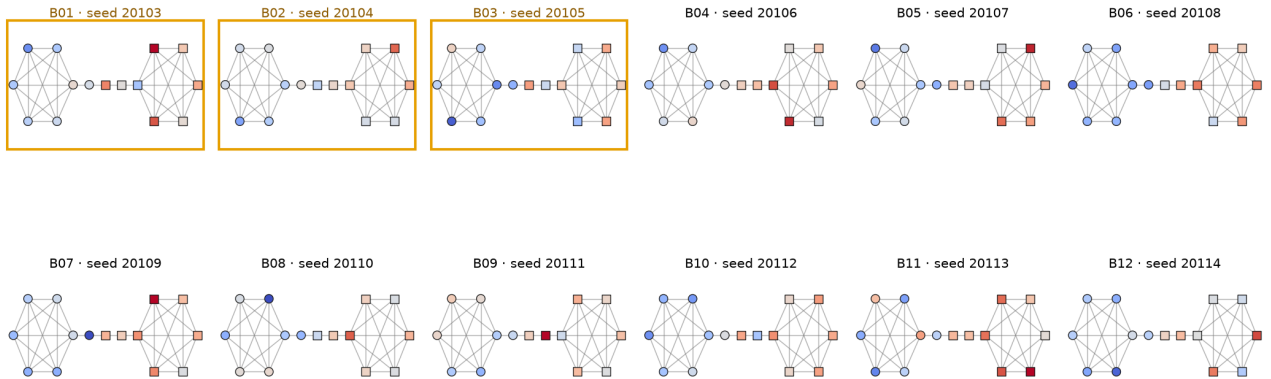


Figure 1: All held-out test graph/signal examples

All three models learned the in-distribution task comparably (mean held-out homophilic accuracy: GCN 0.865; ChebNet 0.861; sheaf 0.878). Under the heterophilic shift, ChebNet scored 0.524, the sheaf baseline 0.469, and GCN 0.312; all were high on the barbell condition (0.964, 0.958 and 0.944 respectively). These accuracy values validate that the models were not merely untrained random maps, but they are not a broad architecture benchmark.

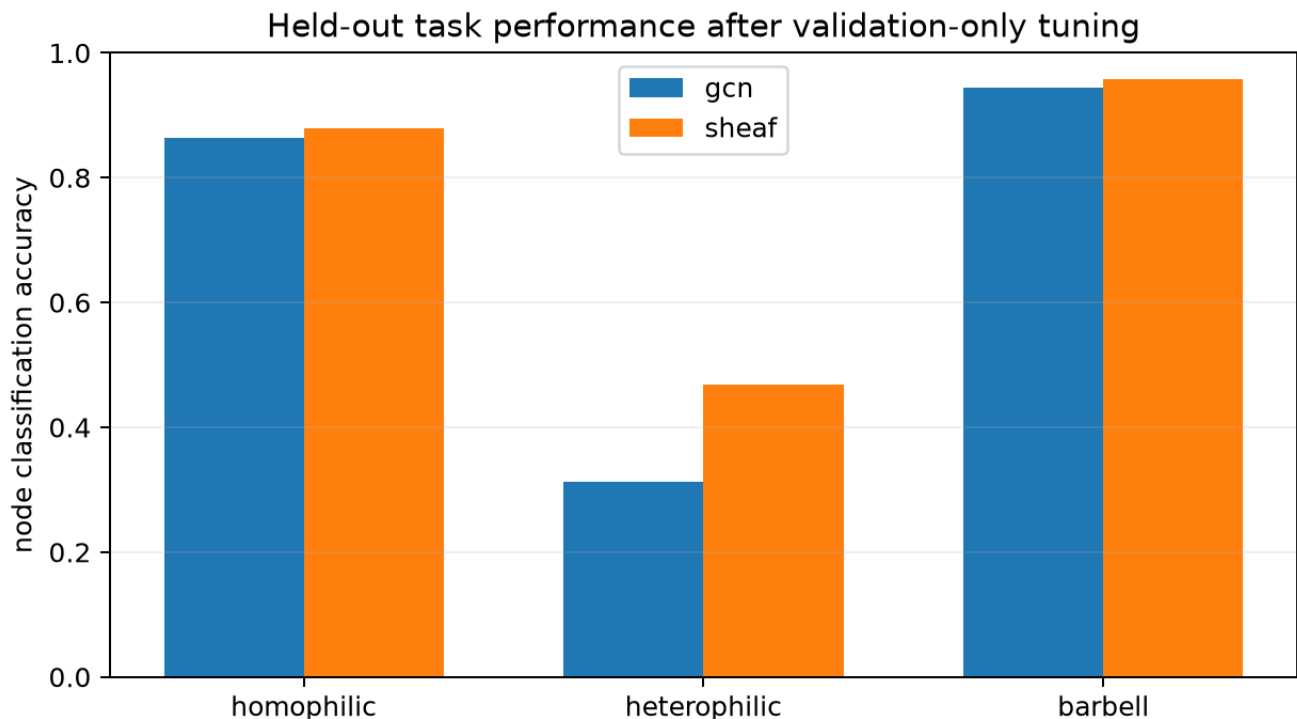


Figure 2: Held-out task accuracy

Verification before measurement

The implementation has Slurm-verified unit tests at clean commit c909dd60d39d51e90ad15413af8470b0da4735fd (job 22347; 8 passed; evidence in RESEARCH_LOG.md):

- explicit linear product equals the JVP-built product;
- zero product returns $-\infty$;
- an actual regular-graph GCN constant mode survives (detecting an accidental P projection);
- non-unit supplied probes are rejected;
- power agrees with dense SVD on a known linear product;
- nonlinear GCN and sheaf JVP products agree with an **independent reverse-mode dense autodiff Jacobian**. The declared ReLU convention is PyTorch’s derivative zero at a kink.

For every small experimental cell, the maximum entrywise discrepancy between JVP-built and reverse-mode product was at most 2.2×10^{-16} for GCN and 2.1×10^{-14} for sheaf. This validates autodiff implementations, but not a hypothetical shared framework bug; it is why both a structural test suite and two AD modes are retained.

Experience 1: random probes converge far too slowly here

The v2 study used 12 independent nested streams through each trained cell, with $M = 1, 2, 4, \dots, 1024$. Shading is the 5th–95th percentile over all selected seeds, trajectories and streams within a graph family.

Random-probe lower-bound convergence by architecture and graph family

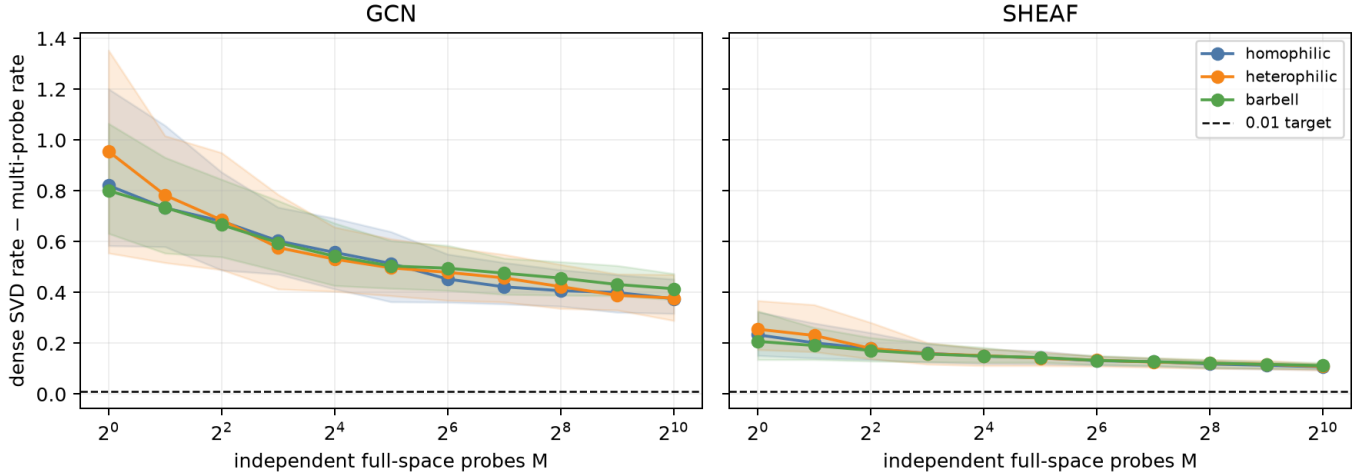
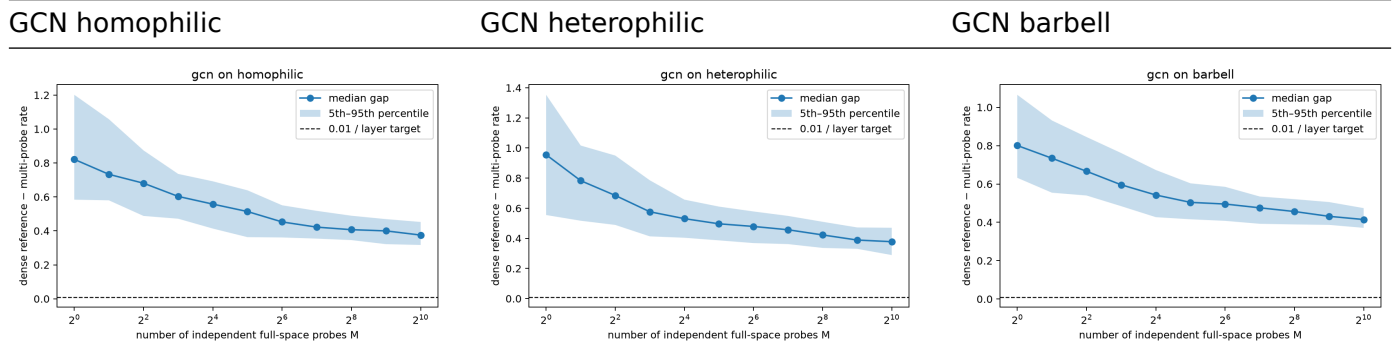


Figure 3: Probe convergence summary

At $M = 1$, pooled median gap was 0.847 (GCN) and 0.231 (sheaf). At $M = 128$, it was 0.449 and 0.127. At $M = 1024$, it was 0.395 and 0.109. The target was 0.01 per layer, so the relevant result is not merely that convergence is slow: **coverage at the target was 0/72 for every model-dataset family at $M = 1024$.**

Model	Dataset family	Median gap at $M = 1024$	95th percentile gap	Target coverage
GCN	homophilic	0.376	0.452	0/72
GCN	heterophilic	0.379	0.469	0/72
GCN	barbell	0.417	0.473	0/72
Sheaf	homophilic	0.107	0.117	0/72
Sheaf	heterophilic	0.109	0.119	0/72
Sheaf	barbell	0.112	0.124	0/72

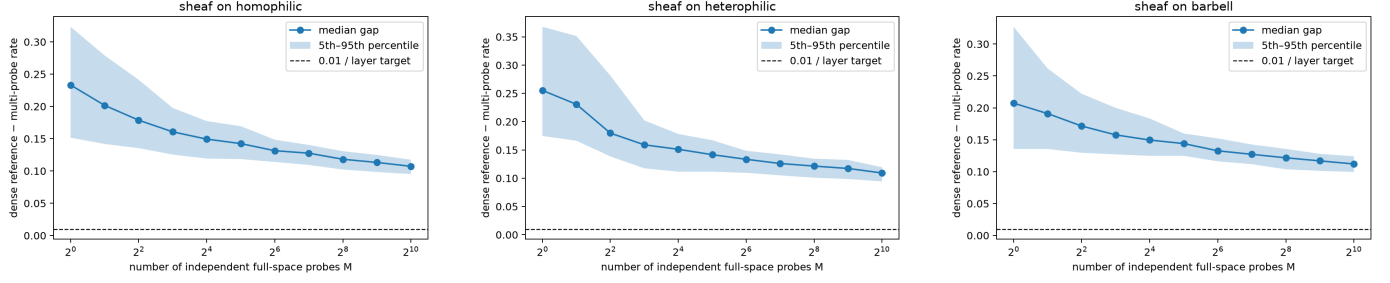
The per-family plots make clear that dataset family changes the curve somewhat, but the architecture-level difference is larger in this pilot:



Sheaf homophilic

Sheaf heterophilic

Sheaf barbell



Interpretation: these data falsify the initially tempting default “use 128 random probes and call the maximum the exponent.” They do not say that a random-probe lower bound is wrong; it is doing exactly what it promises. The high gap suggests that the dominant finite-product direction is fairly concentrated relative to random full-space directions, especially for these trained GCNs and ChebNets.

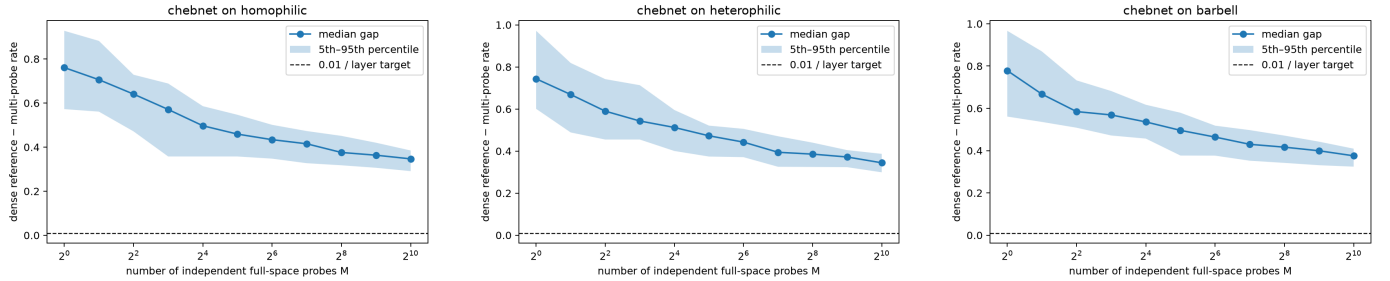
ChebNet replication

A separately tuned depth-3/order-3 ChebNet calibration (two seeds, three trajectories per graph family, 12 streams through 1024 probes) replicates the random-probe conclusion: median $M = 1024$ gaps were 0.347 (homophilic), 0.346 (heterophilic), and 0.376 (barbell), with 0/72 streams at the 0.01 target in every family. All 18 corresponding three-start/160-iteration power cells met the residual criterion. Its curve lies between the GCN and sheaf cases, showing that the result is neither specific to a one-hop filter nor explained only by depth.

ChebNet homophilic

ChebNet heterophilic

ChebNet barbell



Probe convergence through 2^{20} : still far from the target

We pushed the same random-probe estimator to $M = 2^{11}, 2^{12}, \dots, 2^{20}$ independent full-space probes per cell by streaming chunks (two replicates per model/family on the same checkpoints; no infeasibility cutoff was needed—one $2^{20} = 1,048,576$ -probe stream took ~ 9 – 11 s for GCN, ~ 15 – 19 s for ChebNet, and ~ 263 – 329 s for the depth-10 sheaf). Even at a million probes the estimate is still far from the exact reference.

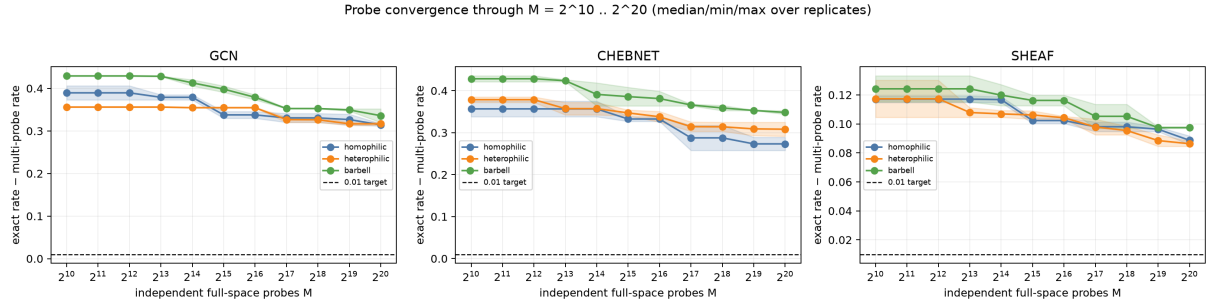


Figure 4: Probe convergence through $M = 2^{20}$

Median gap to the exact reverse-mode reference (pooled over graph families and replicates):

Model	$M = 2^{10}$	$M = 2^{16}$	$M = 2^{20}$	reduction per $1024 \times$ probes
GCN	0.390	0.355	0.318	$\sim 18\%$
ChebNet	0.380	0.344	0.308	$\sim 19\%$
Sheaf	0.117	0.105	0.090	$\sim 23\%$

The estimate **does** converge (it is a rigorous, monotone lower bound that increases with M), but only quasi-logarithmically: multiplying the probe count by 1024 closes at most a quarter of the gap, exactly the overlap-concentration mechanism from the estimator section. Random probes are therefore practically unusable as a point estimator at any feasible budget, now demonstrated over three orders of magnitude of M .

The variation that converges way faster: adaptive (operator-refined) probes

The bottleneck is that independent random probes merely wait for a draw with high overlap on the dominant right-singular direction. The fix is to make the probe depend on the operator instead: start from one unit probe v_0 and iteratively refine it by the symmetric operator $B_K^* B_K$, $v_{m+1} = B_K^* B_K v_m / \|B_K^* B_K v_m\|$ (this is a power step; “ m refinement steps” in what follows), keeping the same log-growth estimate with v replaced by v_m . On the identical seed-90000 cells the two variations were measured back-to-back (run 2026-09-27-probe-vs-adaptive, Slurm 22828):

A probe-estimator variation that converges way faster: refine one probe instead of drawing 10^6

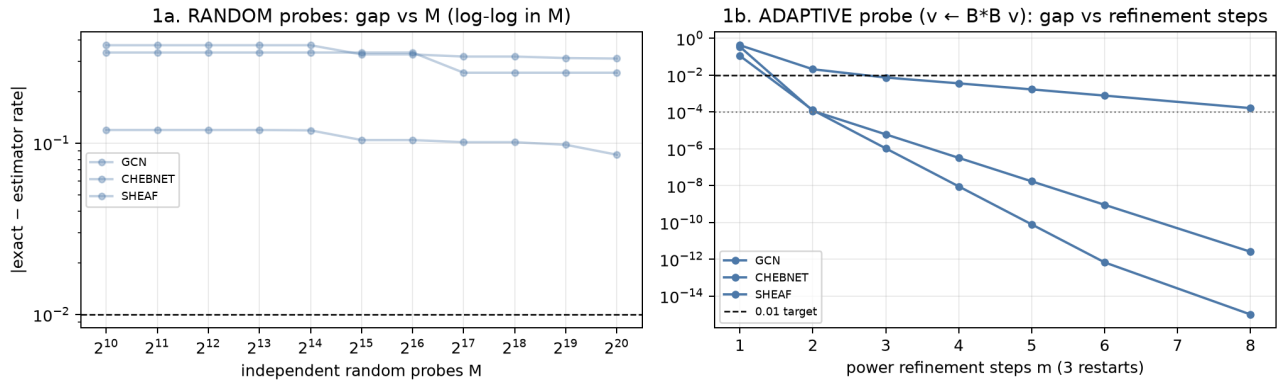


Figure 5: Random vs adaptive probe convergence, same cells

model	random, $M = 2^{20}$	adaptive $m=2$	adaptive $m=3$	adaptive $m=6$
GCN	0.31-0.35	1e-5 - 1.3e-3	6e-6 - 3.1e-5	$\sim 1e-12$
ChebNet	0.26-0.34	1.3e-2 - 2.1e-2	6e-6 - 7.4e-3	1.7e-3 - 1e-11
Sheaf	0.085-0.097	2.4e-5 - 1.8e-4	5e-8 - 1.0e-6	$\sim 1e-14$

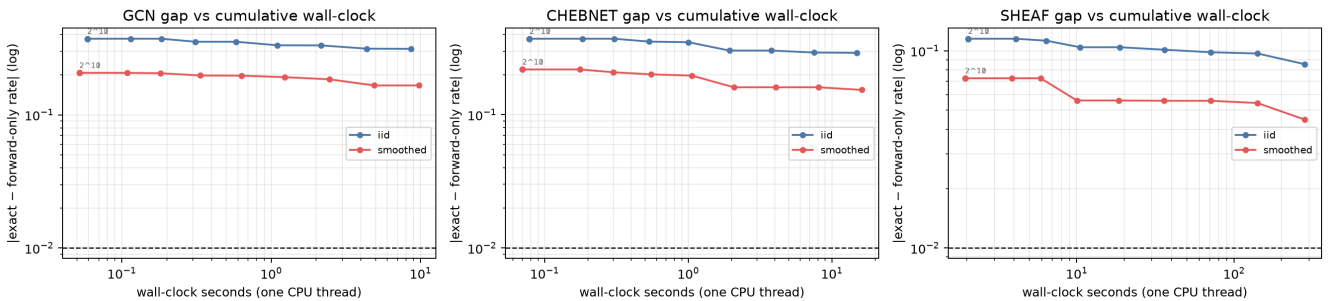
The random estimator stays at a gap of 0.09-0.35 even after a million probes; its fitted law $\text{gap} \approx A - B \log_2 \log_2 M$ ($B = 0.0018-0.010$) extrapolates the budget for a 0.01 gap to $M^* \gtrsim 2^{245}$ (the fitted value overflows double precision, $\approx 10^{(10^{13})}$ probes). The adaptive variation reaches 10^{-4} in **2-4** refinements and 10^{-12} in **5-8**, i.e. linear (geometric) decay in m instead of double-log in M , at a per-refinement $10^{-1}-10^{-4}$ gap factor. Each refinement is one $B_K^* B_K$ product (one forward + one reverse-mode pass), so convergence that the random estimator cannot reach with $\gtrsim 10^{(10^{13})}$ probes is reached with a handful of backprops (wall-clock 0.03-1 s per cell).

One caveat must be kept explicit: the refinement must act on the symmetric $B_K^* B_K$, not on B_K alone. A forward-only single-probe refinement $v \leftarrow B_K v / \|B_K v\|$ converges to the dominant eigenvector of the nonnormal B_K , so its log-growth saturates at $(1/K) \log |\lambda_{\max}(B_K)|$ below the top singular rate by 0.11-0.57 (run 2026-09-27-forward-only-adaptive); the speedup therefore requires reverse mode (the adjoint), which is available whenever gradients are.

The best forward-only estimator (wall-clock): graph-aligned (smoothness-skewed) probes

If the estimator is restricted to **forward-only** (JVP applications of B_K , no adjoint), the adaptive variation is unavailable, and the remaining lever is the probe distribution. Streaming both distributions through the same seed-90000 cells to $M = 2^{20}$ (2026-09-27-forward-only-pareto, Slurm 22846) shows that full-space probes skewed toward smooth graph signals, $q \propto (I + cL)^{-1}z$ for the normalized graph Laplacian L , dominate iid probes on every marker and every cell: the gap at 2^{20} is **$\sim 1.8-2.1\times$ smaller at the same wall-clock** (GCN 0.313 $\rightarrow$ 0.166, ChebNet 0.292 $\rightarrow$ 0.154, sheaf 0.086 $\rightarrow$ 0.045; median wall 9.8/16/281 s). A $-c$ scan at $M=2^{14}$ is monotone (2026-09-27-smooth-c-scan, Slurm 22852): gap GCN 0.373 (iid) $\rightarrow$ 0.091, ChebNet 0.338 $\rightarrow$ 0.077, sheaf 0.119 $\rightarrow$ 0.039 at $c=16$. Marginal batched probe cost is tiny (GCN 8 μ s, ChebNet 14 μ s, sheaf 0.25 ms per probe), so the wall-clock frontier is nearly flat in M ; setup and depth dominate. A frequency-adaptive tilt (reweighting probes toward the nodes that grew fastest, still forward-only) does **not** help: it spends probes learning the tilt and is worse than fixed smoothing at every count (e.g. GCN $M=10$: adapted 0.37 vs smoothed 0.17), so fixed smoothness-aligned probes are the forward-only ceiling: pooled-best gap $\approx$ GCN 0.17, ChebNet 0.21, sheaf 0.08 with even $M=10$ probes, and 0.09/0.09/0.04 at $M=2^{10}$ (run 2026-09-27-forward-adapted). Even the best forward-only lower bound stays $\gtrsim 3\times$ above the 0.01 target at 10^4 probes, confirming the forward-only ceiling is a large constant gap, not a tuning artifact.

Best forward-only estimator wall-clock Pareto (forward probes only; rightmost point = $M = 2^{20}$)



The decisive caveat stands: even the smoothed distribution obeys the same double-log law ($M^*(0.01) \approx 2^{233}-2^{258}$). **The best-possible forward-only estimator in wall-clock is therefore the batched smoothness-aligned full-space multiprobe** — a rigorous monotone lower bound, $\sim 2-5\times$ better than iid at equal wall-clock — and, unlike forward-only refinement (which provably leaves the $\rho(B_K)$ plateau), it keeps the correct $\sigma_1(B_K)$ limit.

Experience 2: power iteration succeeds on the same finite products

For the same 36 trained tiny cells, three random-start power runs of $160 B_K^* B_K$ iterations were performed. The best returned rate was accepted only when its explicit relative residual was $\leq 10^{-7}$. All 36 cells met the condition. Compared with the reverse-mode dense SVD rate, the mean absolute error was 3.9×10^{-17} and maximum 2.2×10^{-16} ; these are tiny-reference agreement results, not an assertion that 160 iterations will suffice at every scale.

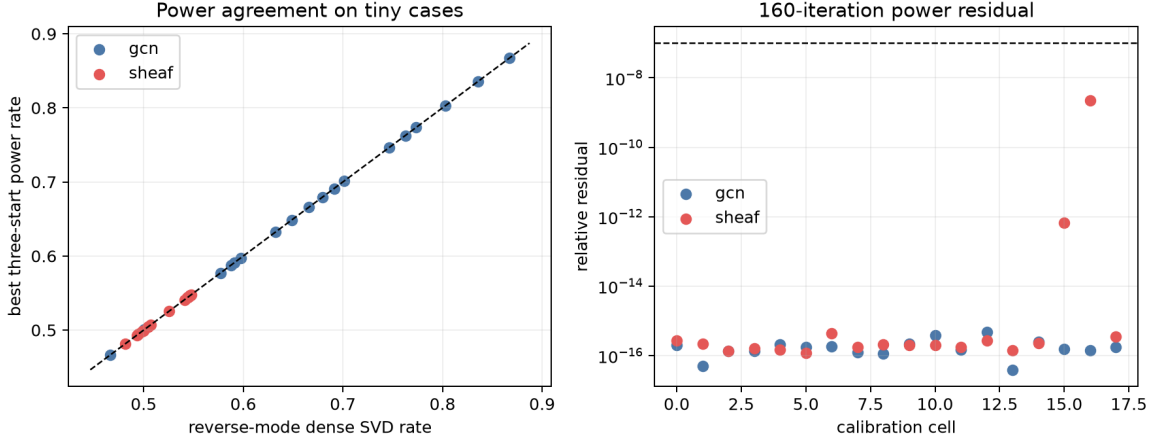


Figure 6: Power versus exact SVD

This contrast is expected: power iteration actively rotates a generic nonzero-overlap start toward the dominant right singular direction, whereas independent raw probes only wait to sample a close one. The final residual is essential. In v1, a single 80-iteration start failed the 10^{-7} residual criterion in two deep-sheaf/barbell cells; v2's three-start/160-iteration protocol resolved all tiny cells. A low residual validates convergence of the chosen Krylov iterate, while agreement of multiple starts helps reveal a poor start or numerical pathology.

Experience 3: rates, graph families and training state

The exact top rate is positive in this pilot. That means selected hidden perturbations can grow across depth; it does not by itself predict prediction accuracy or distant-node sensitivity.

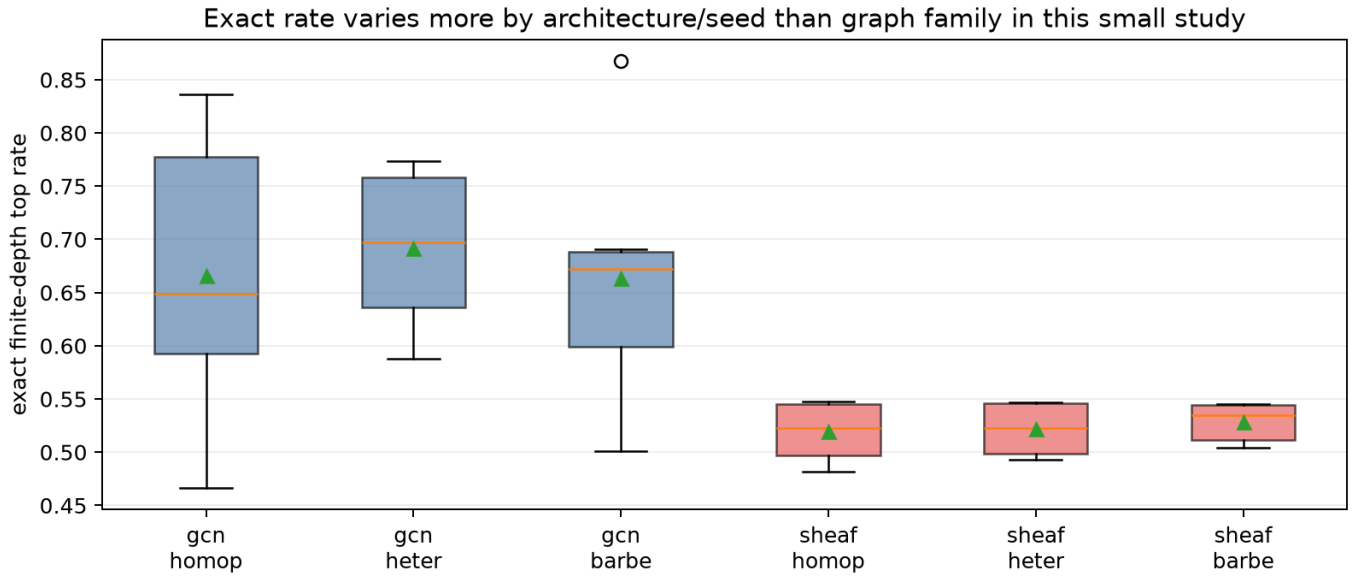


Figure 7: Exact-rate distribution

A separate trained-versus-untrained ablation used two seeds, three trajectories per family, eight streams and up to 512 probes. Training did not make raw probe estimates tight. At $M = 512$, the GCN median gaps ranged from 0.418–0.430 trained and 0.396–0.418 untrained; sheaf gaps ranged from 0.111–0.116 trained and 0.111–0.114 untrained. This is useful negative evidence: random-probe behavior is not safely inferred from whether a model has learned the task.

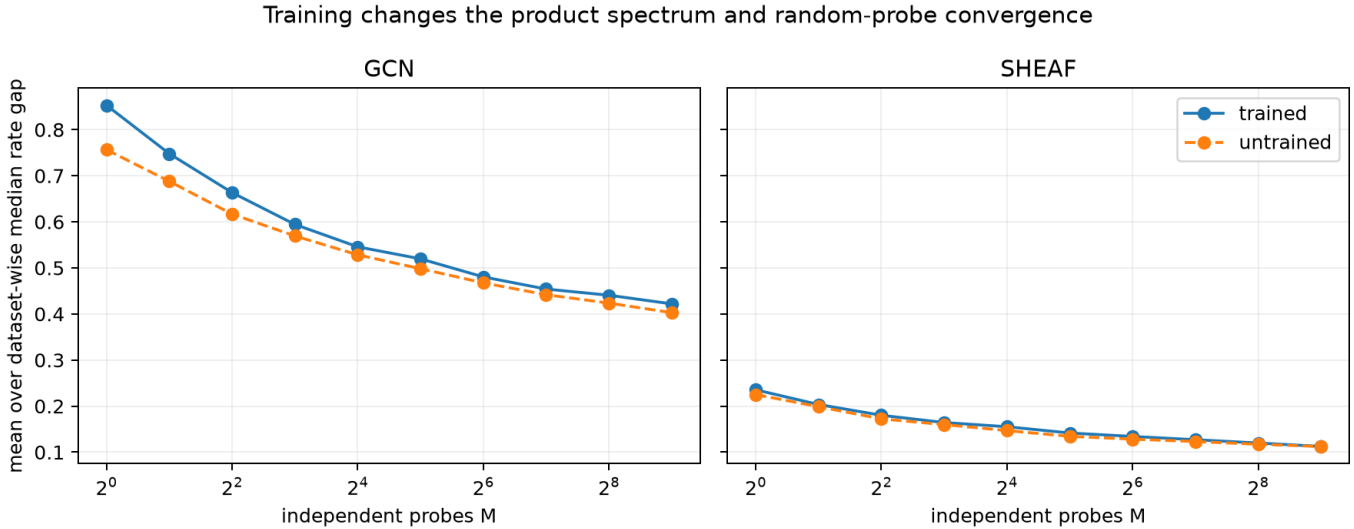


Figure 8: Training-state probe comparison

Experience 4: estimator wall-clock cost

Wall-clock time was measured for methods targeting the same **unprojected** product B_K . The benchmark used the first test trajectory from each graph family, both model seeds, and all three architectures: six fixed (G, x, Φ) cells per architecture. Each method received one warm-up and three timed repeats, giving 18 timings per table entry. PyTorch float64 ran eagerly on one CPU thread in Slurm job 22388. The timed region starts after training, forward-state construction, layer-function setup, and probe generation; it includes only the estimator/solver call.

Median seconds per fixed cell were:

Model	Dense Jacobian + SVD	Probes $M = 128$	Probes $M = 1024$	Power 3×160
GCN	0.0033	0.0038	0.0074	1.690
ChebNet	0.0045	0.0075	0.0138	3.510
Sheaf	0.071	0.103	0.293	38.46

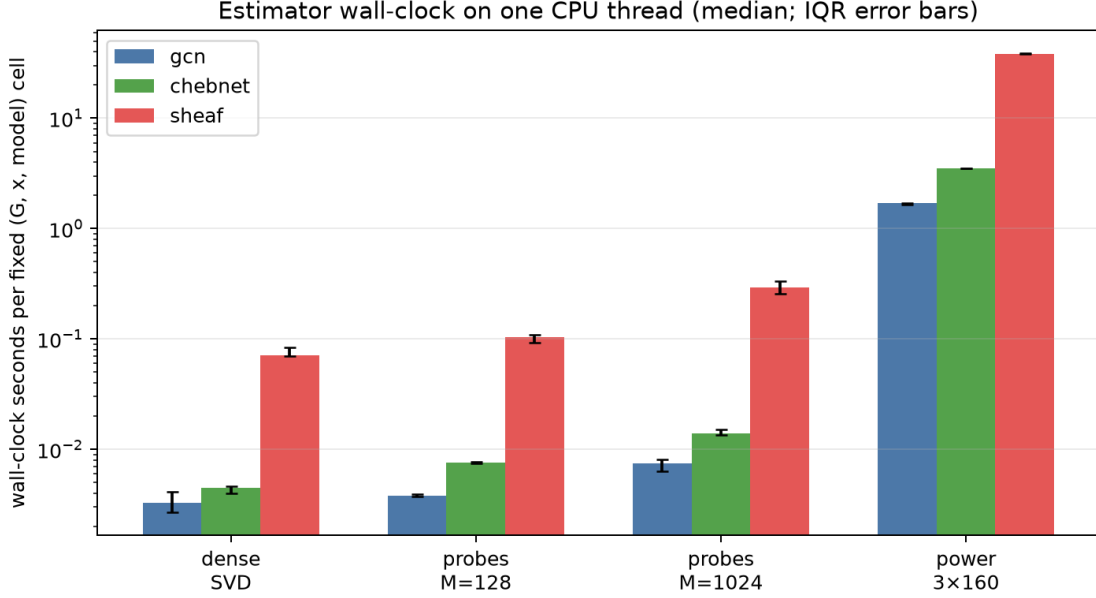


Figure 9: Estimator wall-clock comparison

On these deliberately tiny tangent spaces, dense SVD is both exact and fastest. That is why the recommended protocol uses it for calibration whenever the explicit Jacobian fits. Vectorized random probes are also cheap here, but their severe accuracy gaps remain; low wall-clock cost does not make them good point estimates. The residual-checked power protocol is much slower in this eager CPU implementation because three starts at 160 iterations require 480 forward/adjoint product iterations, especially through the depth-10 state-adaptive sheaf. All 54 timed power repeats converged, with maximum relative residual 4.0×10^{-12} .

This timing result does **not** overturn the scalable recommendation. Dense Jacobian storage grows quadratically with tangent dimension and eventually becomes infeasible, whereas matrix-free power avoids materializing B_K . Conversely, the vectorized $M = 1024$ timing uses memory proportional to the probe batch. Absolute timings and crossover points will change with dimension, hardware, compilation, batching, and a Lanczos implementation; this is a transparent small-CPU benchmark, not a universal speed ranking.

Significantly faster matrix-free estimation: early-stopping power and Lanczos

Two cheap upgrades cut the matrix-free cost dramatically on the same cells (same checkpoints, nine cells per model, one warm-up and three timed repeats each; one CPU thread):

- **Early-stopping power:** run each start's power iteration but stop as soon as its residual meets $\leq 10^{-7}$ (checked every 10 sweeps) instead of always running the full 160, so the cost is only the converged prefix.
- **Matrix-free Lanczos:** one Lanczos run on $B_K^* B_K$ with full reorthogonalization; the largest Ritz value gives the top singular value with a directly checkable Ritz residual.

Median seconds per cell and speedup over the fixed 3×160 protocol:

Model	Power 3\$×160 $Power_{early}$	Lanczos	Speedup (early)	Speedup (Lanczos)	
GCN	1.26	0.086	0.12	14.7×	10.4×
ChebNet	2.64	0.36	0.49	7.3×	5.4×
Sheaf	30.7	2.06	5.9	14.9×	5.2×

Every one of the 27 cells reached residual $\leq 10^{-7}$ (power early) and every Lanczos run converged, matching the exact rate to numerical precision (median gap $\leq 10^{-16}$). The speedup is largest exactly where the baseline was worst: the depth-10 sheaf’s power cost drops from ~ 31 s to ~ 2.1 s with early stopping. Full per-cell data is in evidence/2026-09-26-power-speedup/. Both routines are now part of the `lyapunov_diagnostics.estimators` module (`power_rate(check_every=...)` and `lanczos_top_rate()`).

A loose 10^{-2} residual helps only if the check is fine-grained. The residual is compared at most once per `check_every` sweeps, and on these tiny matrices power gets below 10^{-7} by sweep 10—so with `check_every=10` the loose threshold looks identical (10^{-7} 0.083/0.35/2.00 s vs 10^{-2} 0.082/0.17/2.00 s). Checking every sweep reveals the true gain: the residual falls below 10^{-2} by sweep 3, giving (median seconds, evidence 2026-09-27-power-speedup-loose-fine):

Model | 3\$×160 | early 10^{-7} (chk 10) | early 10^{-2} (chk 1) | speedup vs 3\$×160 | $\max\|rate - exact\|$ |
|—|—|—|—|—|—| GCN | 1.21 s | 0.083 s | 0.045 s | **26.8×** | 3×10^{-5} | | ChebNet | 2.54 s | 0.347 s |
0.136 s | **18.7×** | 2.5×10^{-4} | | Sheaf | 29.3 s | 1.99 s | 1.09 s | **26.8×** | 2.4×10^{-5} |

So the loose threshold can roughly **double** the early-stop speedup (to ~ 19 – $27\times$) when checked every sweep, at the cost of a worst-case rate error $\sim 10^{-4}$; with a coarse check schedule the benefit is hidden because the tight threshold already passes the first check.

Experience 5: exact Lyapunov-rate histograms across models, datasets, and heat diffusion

For each GNN we load a trained checkpoint Φ (depth-3 GCN, depth-3/order-3 ChebNet, depth-10 sheaf; each 0.986 validation accuracy) and draw **the same** 300 graph/signal pairs per family (one shared (G, x) distribution, seeds offset identically for all three models). For each pair the exact reverse-mode dense Jacobian+SVD rate $\lambda_K(G, x; \Phi)$ is computed, giving 900 pointwise samples per model. We also evaluate the same draws under a fourth, **non-GNN** system: linear graph heat diffusion $\dot{x} = -Lx$, discretized with a stable Euler step $M = I - dt L$ over depth 10 and the symmetric normalized Laplacian. Its empirical distributions are plotted together below.

Exact top (1st) eigenmode rates over random (G, x) pairs (same 300 draws/family)

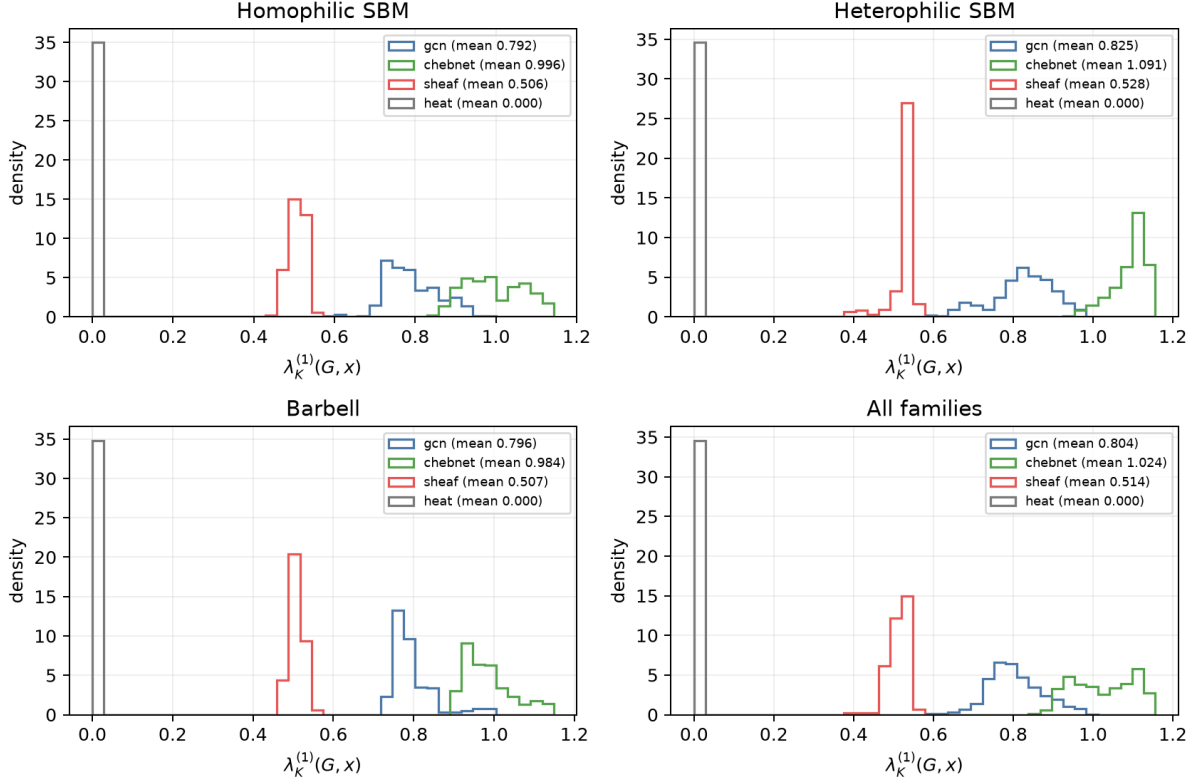


Figure 10: Exact Lyapunov-rate histograms over random (G, x) pairs

Mean exact rates (with median and standard deviation) over the 900 draws per system:

System	Overall mean	Overall median	Std	Homophilic	Heterophilic	Barbell
GCN	0.804	0.793	0.068	0.792	0.825	0.796
ChebNet	1.024	1.028	0.076	0.996	1.091	0.984
Sheaf	0.514	0.517	0.026	0.506	0.528	0.507
Heat diffusion	≈ 0	≈ 0	$\sim 10^{-17}$	≈ 0	≈ 0	≈ 0

Four features stand out.

1. **Model ordering is clean and consistent.** ChebNet dominates every family (mean ≈ 0.98 -1.09), GCN is intermediate (≈ 0.79 -0.83), and the deep sheaf baseline is lowest (≈ 0.51 -0.53). The histograms are well separated, so a single top rate meaningfully differentiates these trained architectures rather than being buried in sampling noise.
2. **The distributions are narrow relative to their means** (coefficient of variation $\lesssim 0.09$). Across hundreds of graph/signal draws, $\lambda_K(G, x; \Phi)$ varies by only ~ 0.05 -0.08 (one standard deviation), so the finite-depth growth rate is fairly insensitive to which specific graph/signal input is chosen within a fixed model—at least on this synthetic family scale.
3. **Family has a smaller effect than model.** Within a model, family means sit within ~ 0.03 for the sheaf and GCN and ~ 0.1 for ChebNet, whereas model-to-model differences exceed 0.29 everywhere. The heterophilic condition tends to raise the mean for all three, consistent with the earlier pointwise calibration; it does not reorder the models.
4. **Heat diffusion is a clean negative control.** Its top rate is exactly 0 (mean 2.7×10^{-17}): the graph Laplacian preserves the constant node mode and decays every other direction, so $\sigma_1(I - dt L) = 1$

and $\lambda = 0$ for every connected draw. The exact estimator recovers this to numerical precision, directly demonstrating that the positive GNN rates above are genuine amplification rather than an artifact of the estimator. It is mass-preserving and non-expanding, unlike the trained networks.

As throughout this report, these are **finite-depth** rates under the fixed-draw empirical distribution, computed with the exact reference; they are not infinite-depth asymptotic Lyapunov exponents and say nothing alone about oversquashing or task utility.

Modes 1, 13, and 25

The exact estimator returns the full singular spectrum of B_K , so each draw gives eigenmode rates $\lambda_K^{(r)} = (1/K) \log \sigma_r(B_K)$ for $r = 1, \dots, 25$. The per-system panel below overlays three widely spaced modes for the GNNs—1, 13, and 25—using the same colour and a solid/dashed/dotted line. Heat (single scalar field) has too few modes for 13/25, so its panel keeps modes 1/2/3.

Eigenmode rate histograms per system (GNNs: modes 1/13/25; heat: modes 1/2/3)

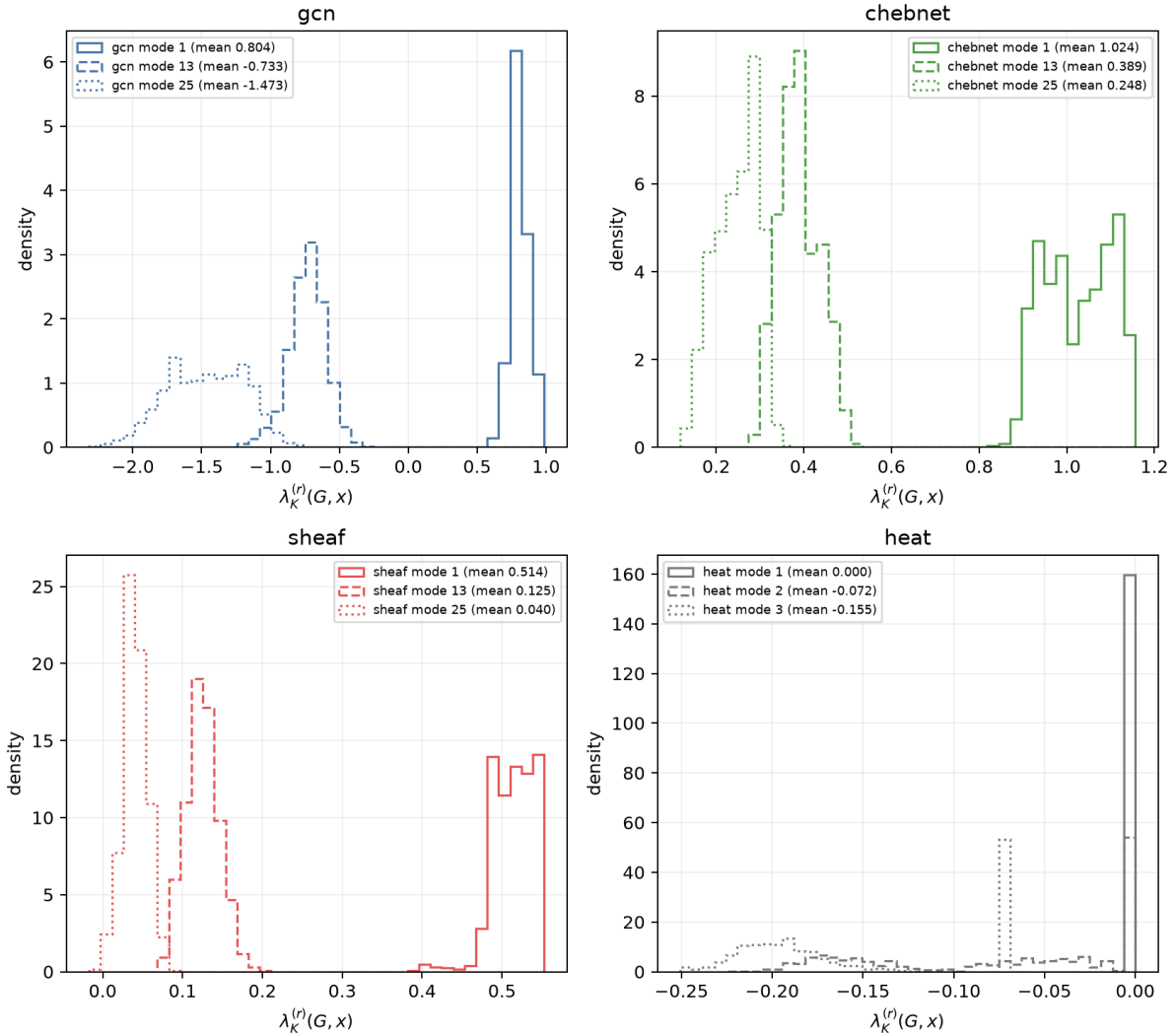


Figure 11: Finite-depth eigenmode rate histograms per system

Mean eigenmode rates over the 900 draws per system (GNNs at modes 1/13/25; heat at its modes 1/2/3):

System	$\lambda^{(1)}$	$\lambda^{(13)} / \lambda^{(2)}$	$\lambda^{(25)} / \lambda^{(3)}$	Total drop
GCN	0.804	−0.733	−1.473	2.277
ChebNet	1.024	0.389	0.248	0.776
Sheaf	0.514	0.125	0.040	0.473
Heat	≈ 0	−0.072	−0.155	0.155

For ChebNet and the sheaf, the decay tapers slowly and remains positive through mode 25. For GCN the spectrum plunges below zero by mode 13 (mean -0.73) and reaches -1.47 by mode 25: growth is concentrated in just the leading few GCN directions, while the vast majority of hidden directions are genuinely contractive. Heat shows no such plateau—mode 1 stays 0 (constant mode preserved) and modes 2/3 are negative (contraction), exactly the physics of $\dot{x} = -Lx$ and in stark contrast to the positive GNN spectra.

Separate experiment: Pierre’s projected estimator

Pierre’s proposal removes the node-constant component of every tangent after every layer. Let

$$Pz = z - \frac{1}{N} \mathbf{1}\mathbf{1}^\top z$$

act independently on each feature channel. Mean-zero unit probes are initialized in $\mathbf{1}_N^\perp$, and the measured finite product is

$$B_K^P = PJ_{K-1}PJ_{K-2}\cdots PJ_0|_{\mathbf{1}_N^\perp}.$$

This is a coherent transverse/mean-zero observable, but it is **not an estimator of** the actual hidden-state Jacobian $B_K = J_{K-1}\cdots J_0$: projection removes any path that enters a constant node mode and later leaves it. The caveat is stronger for normalized graph operators and sheaves, whose natural null or harmonic subspaces need not equal the raw node-mean-zero subspace. The experiment implements Pierre’s literal node-mean prescription and reports it under a distinct name.

The projected study repeated the same design with matched data/model seeds and tuning grid for all three architectures: 18 pointwise cells per architecture, 12 nested mean-zero streams through $M = 1024$, a dense SVD of the reduced projected product, and three-start/160-iteration projected power iteration. All **54** projected power cells passed relative residual $\leq 10^{-7}$ and agreed with projected dense SVD to numerical precision.

Mean exact rates over the 18 matched cells per architecture were:

Model	Full Jacobian B_K	Pierre projected B_K^P	Reduction
GCN	0.674	0.344	0.330
Sheaf	0.523	0.427	0.096
ChebNet	1.112	0.844	0.268

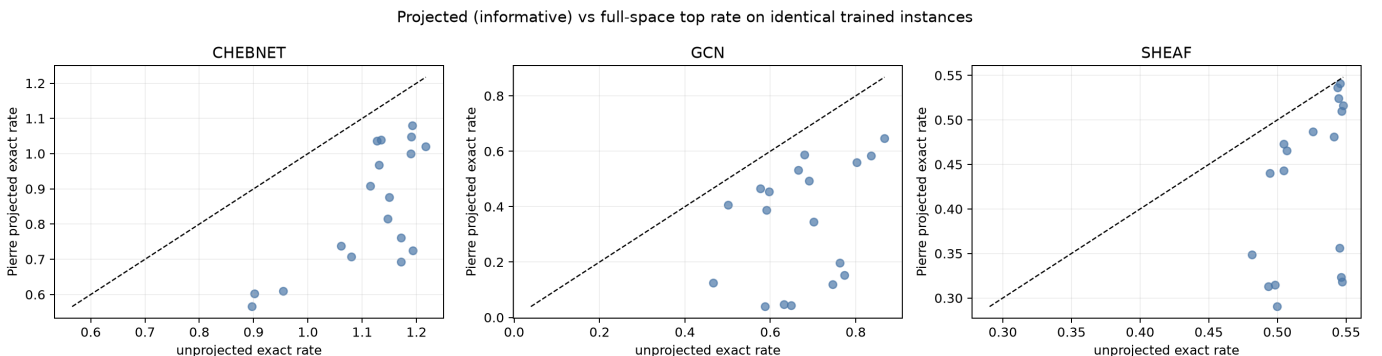


Figure 12: Projected versus unprojected exact rates

Projection therefore lowers the architecture-mean rate in every case, most strongly for GCN. This supports the interpretation that consensus-like directions elevate the full-space top rate. It does **not** make the projected rate generally negative: the projected exact means remain positive, and ChebNet retains substantial non-constant amplification.

The random-probe difficulty also remains. At $M = 1024$, no projected stream reached the 0.01-per-layer target. Comparing the actual finite- M averages—not the exact references—over $18 \text{ cells} \times 12 \text{ streams}$ gives:

Model	Full-space probes	Pierre probes	Pierre minus full
GCN	0.280	-0.038	-0.318
Sheaf	0.414	0.324	-0.090
ChebNet	0.759	0.517	-0.242

These $M = 1024$ numbers are lower-bound estimates and should not replace the exact/power comparison above. For example, the negative average projected GCN probe result does not establish that its true projected top rate is negative; the projected exact mean is 0.344. It shows how severely random directions can miss the maximizing direction even after mean-zero projection.

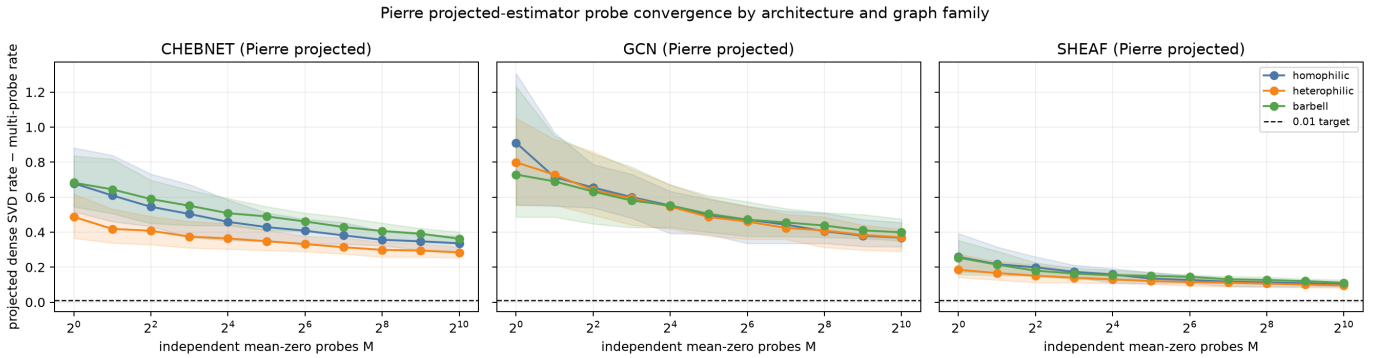


Figure 13: Pierre projected-probe convergence

Full cell-level evidence and caveats are in `notes/pierre-estimator-experiments.md`.

Recommended default protocol

Situation	Recommendation	Required reporting
Tiny calibration, e.g. $D \lesssim 200$	Reverse-mode dense Jacobian/SVD plus JVP-product agreement	Both AD routes, max product discrepancy, exact singular value/rate
Routine finite- K top rate with JVP and VJP	Early-stopping multi-start power on $B_K^* B_K$ ($\text{power}_{\text{rate}}(\text{check}_{\text{every}} > 0)$); accept only residual $\leq 10^{-7}$; Lanczos when available — 5-15$\times$ faster than the fixed 3$\times$160 protocol	each start's rate/residual, convergence iterations, accepted start, JVP/VJP sweep count, precision
Power fails residual/start agreement	Run more iterations; then matrix-free Lanczos/Krylov; otherwise label result unresolved	residual history and stop reason; never silently emit a number

Situation	Recommendation	Required reporting
JVP-only setting	Report nested multi-probe maxima explicitly as lower bounds; calibrate $\bar{M}$ against a trusted reference by architecture/dimension	$\bar{M}$, stream count/seeds, gap coverage curve; no unqualified “exponent” claim
Long, tied/stationary depth process and spectrum needed	QR/Benettin frame, with an explicitly justified stationary cocycle	transient/renormalization schedule and convergence history
Any GNN	evaluation mode; fixed graph and stochastic masks; full derivative through adaptive geometry/normalization; no P	state boundary, hidden width/depth, dtype, graph/example IDs, full commit and lock hashes

The 160/three-start choice is an evidence-supported **initial default**, not a permanent magic number. At larger dimensions, the residual threshold and repeat-start checks—not the iteration count—should decide acceptance. Lanczos is the next algorithmic upgrade because it uses the same JVP/VJP primitives but can improve convergence for an unfavorable spectrum.

Limits and next experiments

- The graphs are deliberately small because exact SVD is central to calibration. We have not established a power-iteration residual threshold on large real-world GNNs.
- The sheaf architecture is an explicit diagnostic baseline, not a faithful reproduction claim for every Neural Sheaf Diffusion variant. The state-dependent- μ GNN remains underspecified in notes/state-dependent-mu-diffusion-GNN.md.
- The study is node classification on synthetic graph distributions; it does not yet cover citation/coauthor/molecular benchmarks, graph-level tasks, dropout/DropEdge or heterogeneous feature sizes.
- The calibration varies (G, x) jointly and has only three examples per graph condition and model seed. It therefore does not estimate input-only variation $x \mapsto \lambda_K(G, x; \Phi)$ on a fixed graph, nor a high-precision population distribution.
- Global $\sigma_1(B_K)$ cannot diagnose particular source-target blocks of the Jacobian, delocalization, over-smoothing or oversquashing. Those need separate controlled observables.
- The next scale-up should retain this exact-SVD calibration suite, then test power/Lanczos residual behavior and rate distributions on tuned GCN, ChebNet, a specified sheaf implementation, and the specified adaptive- μ model on real benchmarks. Hyperparameters must be selected from validation data only.

Evidence and reproducibility

Raw outputs and tracked evidence use results/<run-id>/ and evidence/<run-id>/ respectively. Full hashes and paths are in the manifests and RESEARCH_LOG.md.

- **v1** (up to 128 probes): run ID 2026-09-24-probe-calibration-v1; commit e33f696; Slurm 22348.
- **v2** (up to 1024 probes; 3×160 power): run ID 2026-09-24-probe-calibration-v2; commit 12bed935; Slurm 22351.
- **Initialization ablation:** run ID 2026-09-24-probe-calibration-initialization; commit 151934d; Slurm 22355.
- **ChebNet replication:** run ID 2026-09-24-probe-calibration-chebnet; commit 1224c6b; Slurm 22359.
- **Pierre projected estimator:** run ID 2026-09-24-pierre-estimator-calibration; calibration commit 3aab845; Slurm 22377; curation Slurm 22380.
- **Estimator wall-clock:** run ID 2026-09-24-estimator-wall-clock; commit 8384aec; validation Slurm 22387; benchmark Slurm 22388.
- **Test-dataset overview:** run ID 2026-09-24-test-dataset-overview; final plotting Slurm 22390.

- **Lyapunov-rate histograms:** per-model runs 2026-09-24-lyapunov-histogram-{gcn, chebnet, sheaf, heat} (900 draws each, Slurm 22400/22406) compared by curation (Slurm 22402/22407) into evidence/2026-09-24-lyapunov-histogram-comparison/. GNN checkpoints trained offline (jobs 22398/22399) under checkpoints/; heat diffusion is an unlearned system (no checkpoint).
- **Probe convergence through 2^{20} :** run ID 2026-09-26-probe-convergence-2p20; Slurm 22803 (CPU 2 / 16G) streamed independent probes to $M = 2^{20}$ with no cutoff (GCN $\sim 9\text{--}11$ s, ChebNet $\sim 15\text{--}19$ s, sheaf $\sim 263\text{--}329$ s per stream); all 6 streams per model reached 2^{20} . Median gaps at $2^{10\rightarrow 20}$: GCN 0.390 $\rightarrow$ 0.318, ChebNet 0.380 $\rightarrow$ 0.308, sheaf 0.117 $\rightarrow$ 0.090.
- **Adaptive (operator-refined) probe variation (goal answer):** run 2026-09-27-probe-vs-adaptive; Slurm 22828 (CPU 2 / 16G) measured random-probes-to- 2^{20} AND BB-refinement $m=1..8$ on identical seed-90000 cells: gap $\leq 1e-4$ in 2-4 refinements, $1e-12$ in 5-8 (GCN 0.31-0.35 at 2^{20} ; ChebNet 0.26-0.34; sheaf 0.085-0.097); random law $A - B \log_2 \log_2 M$ ($B=0.0018\text{--}0.010$) extrapolates $M(0.01) \approx 2^{(245)}$ (overflows double). Follow-up 2026-09-27-forward-only-adaptive (Slurm 22829) shows forward-only refinement ($v \leftarrow Bv/\|Bv\|$) saturates at $(1/K)\log|\lambda_{\max}(B)|$, gap 0.11-0.57: the transpose (backprop) variant is the required variation.
- **Best forward-only estimator (wall-clock):** 2026-09-27-forward-only-pareto (Slurm 22846) streamed iid vs graph-smoothness-aligned probes ($q \propto (I + cL)^{-1}z$) through 2^{20} per cell: smoothed gap@ 2^{20} GCN 0.166 vs 0.313, ChebNet 0.154 vs 0.292, sheaf 0.045 vs 0.086 at equal wall (9.8/16/281 s); monotone c -scan 2026-09-27-smooth-c-scan (Slurm 22852) gap@ 2^{14} GCN 0.373 $\rightarrow$ 0.091, ChebNet 0.338 $\rightarrow$ 0.077, sheaf 0.119 $\rightarrow$ 0.039 at $c=16$; batched marginal cost $8e-6/1.4e-5/2.5e-4$ s/probe; still double-log $M^*(0.01) \approx 2^{233..58}$. Best forward-only = batched smoothed full-space multiprobe.
- **Power-iteration speedup:** run ID 2026-09-26-power-speedup (Slurm 22796/22803) times fixed 3×160 power vs early-stopping power (power_rate(check_every=10)) vs matrix-free Lanczos (lanczos_top_rate); median speeds sheaf 30.7 $\rightarrow$ 2.06 s (early, **14.9x**) and 5.9 s (Lanczos, **5.2x**), all 27 cells converged $\leq 1e-7$.

All runs used Python 3.12.3 and PyTorch 2.14.0+cu130 in float64.

Lockfile SHA-256: f1fc70f6db931bcdfffb94d353d7ff9e70ed1bcb93899b0efde5668599f7c470e. Curation jobs were 22352/22353/22356. Full manifests state seeds, selected hyperparameters, tensor dimensions, training metrics, power residuals and environment. RESEARCH_LOG.md holds command/log paths and prior validation evidence.

References

Local-source provenance is indexed in papers/README.md.

- Kipf & Welling, Semi-Supervised Classification with Graph Convolutional Networks, arXiv:1609.02907.
- Defferrard, Bresson & Vandergheynst, Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering, arXiv:1606.09375.
- Bodnar et al., Neural Sheaf Diffusion, arXiv:2202.04579.
- Oono & Suzuki, Graph Neural Networks Exponentially Lose Expressive Power for Node Classification, arXiv:1905.10947. Its contraction result concerns distance to an invariant/smoothed subspace under assumptions; it is not a license to insert P into our estimator.
- Johansson, Strannegård, Engsner & Mostad, Exact Spectral Norm Regularization for Neural Networks, arXiv:2206.13581. Matrix-free JVP/VJP power route for a Jacobian spectral norm.
- Cui, Yan, Liu & Lu, Generalizing and Improving Jacobian and Hessian Regularization, arXiv:2212.00311. Matrix-free Lanczos motivation for Jacobian extremal norms.
- Kuczyński & Woźniakowski, Estimating the Largest Eigenvalue by the Power and Lanczos Algorithms with a Random Start, SIAM J. Matrix Anal. Appl. 13(4), 1992, DOI:10.1137/0613066. Random-start and Krylov convergence dependence.
- Benettin et al., Lyapunov Characteristic Exponents for Smooth Dynamical Systems, Meccanica 15, 1980. QR/orthonormal-frame context for asymptotic spectra.